

EBN111 SEMESTER TOETS/TEST 1 (14 MAART/MARCH 2013)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	MEMO	Studentenommer: Student number:							
Voorletters / Initials:		Tel. nr. / Tel no.:							
Dissipline / Discipline:		Taal / Language:				Afr	Eng		

Q1		Q2		Q3		Tot:	/50
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VRAAG 1 – QUESTION 1 (11)				
1.1	$P = 20 \text{ W}$	$W = 2.1 \text{ kWh}$	$W = 7560 \text{ kJ}$	$i(4s) = -25 \text{ A}$
1.2	$v_1 = 8 \text{ V}$	$v_2 = 30 \text{ V}$	$i_1 = -1 \text{ A}$	
1.3	$P_{(6A)} = -100 \text{ W}$	$P_{(R2)} = 100 \text{ W}$	$R_1 = 16.67 \Omega$	

VRAAG 2 – QUESTION 2 (25)			
2.1	a) Nodusse / Nodes = 3		
	b) Onafhanklike lusse: Independent loops: 4	Takke: Branches: 12	Nodusse: Nodes: 9
2.2	$V_0 = -3 \text{ V}$	$P = -100 \text{ W}$	
2.3	$V_{ab} = -35 \text{ V}$	$I_1 = 500 \mu\text{A}$	
2.4	a) $V_Z = -1 \text{ V}$	b) $I_Z = 4 \text{ A}$	
2.5	a) $V_X = 3 \text{ V}$	b) $I_X = -1.5 \text{ A}$	
2.6	a) $V_{Y1} = 20 \text{ V}$	$V_{Y2} = 30 \text{ V}$	
	b) $I_{Y1} = -3 \text{ A}$	$I_{Y2} = 13 \text{ A}$	
2.7	Circuit (1): $R_{eq} = 40 \Omega$	Circuit (2): $R_{eq} = 4 \text{ k}\Omega$	Circuit (3): $R_{eq} = 30 \Omega$

VRAAG 3 – QUESTION 3 (14)

3.1	$a = 1$	$b = -1$
	$c = -2$	$d = 3$
3.2	$e = -6$	$f = -50$
	$g = 8$	$h = 5$
3.3	$V_A = 37.5 \text{ V}$	
3.4	$V_I = -1.2 \text{ V}$	
	$i_3 = 1 \text{ A}$	